

SwachhhLens

Technical Documentation

AI-Powered Waste Response Decision Support System

WORKING PROTOTYPE

HUMAN-IN-THE-LOOP

NOT
PRODUCTION-DEPLOYED

Submitted for

TechNova: Igniting Brilliance (Season 3) — Problem Statement 2, organized by ESSPL

Team TechTitans

Md Farhan (Team Leader) · Ayush Kumar Chaudhary · Junaid Alam

Guru Nanak Institute of Technology, Kolkata — B.Tech Computer Science & Engineering, 3rd Year

Verified implementation: Google Gemini 3.6 Flash (gemini-3.6-flash) · Firebase Authentication
· Cloud Firestore · React/Vite · Leaflet/OpenStreetMap

Purpose: Current prototype architecture, decision logic, assumptions, compliance status, privacy controls and limitations. Implemented capabilities are deliberately separated from future production improvements.

Executive Summary

SwachhLens is a two-application decision-support system for municipal solid-waste complaints. The citizen app captures an image, location, timestamp and optional description; client-side processing prepares the evidence and derives duplicate, priority and intervention signals. Google Gemini 3.6 Flash provides multimodal classification, while deterministic rules keep the operational logic inspectable.

The municipal portal turns those signals into an operational work queue. A municipal operator reviews the AI output, accepts or overrides the recommendation, and dispatches a team. A field supervisor then executes the assigned job and submits completion evidence; the operator performs the final before/after verification. The citizen can review the verified result and submit feedback or request reopening.

Core design principle: AI recommends; authorized people decide, execute, verify and close.

Layer	Primary responsibility
Citizen	Report, track and review
AI + decision engines	Classify, prioritize, detect duplicates and recommend intervention
Municipal operator	Review, dispatch, verify and rework
Field supervisor	Execute assigned work and submit evidence
Citizen closure	Review verified evidence and provide feedback

Problem & Solution Fit

The core problem is operational: a report becomes useful only when a municipality can decide what matters first, which resources fit the incident, where work is concentrated, and whether the cleanup actually occurred.

Operational chain: Report → Understand → Prioritize → Route → Execute → Verify → Close

Problem Statement & Solution

Urban waste reporting is commonly reactive. A citizen reports a problem, an official inspects it, and only then does the municipality decide what manpower, vehicle or response team is appropriate. At intake, several questions remain open: what type of waste is present, how much is present, whether the report is a duplicate, which reports are urgent, and how limited resources should be prioritized.

SwachhLens converts that raw report into structured operational intelligence while keeping consequential decisions in human hands.

01 Capture — Photo + GPS + timestamp + optional description

02 Understand — Gemini 3.6 Flash returns waste type, volume, confidence, sensitivity and bio-risk

03 Prioritize — Duplicate evidence + explainable priority + urgent escalation

04 Route — Rule-based intervention + workload-aware team recommendation

05 Execute — Field supervisor tracks arrival, work and completion evidence

06 Verify & Close — Municipal before/after verification → citizen review

Architecture & Data Flow

Both applications share Firebase Authentication and Cloud Firestore. Gemini provides external multimodal perception; deterministic decision engines run in the browser; the municipal portal provides the human approval points. The current prototype intentionally avoids Cloud Functions, Cloud Run and Firebase Storage.

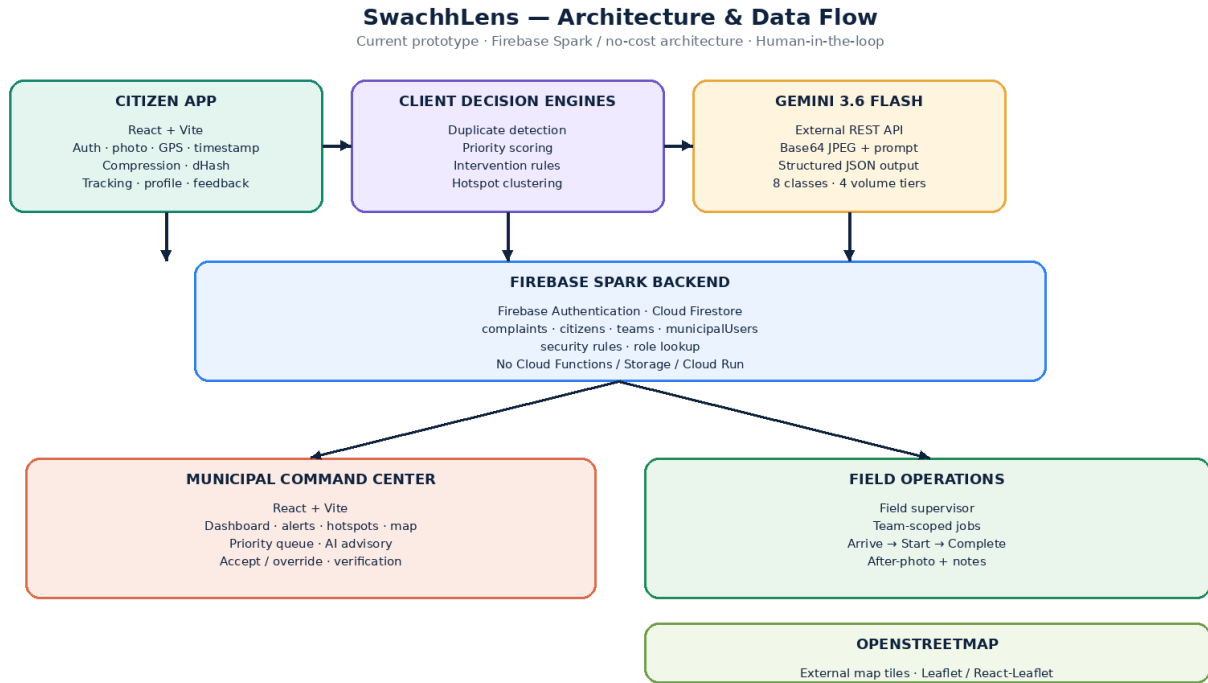
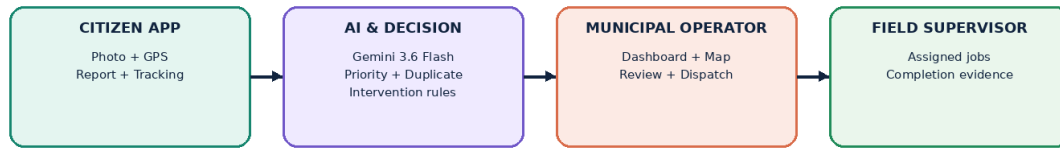


Figure 1. Detailed current-prototype architecture and data flow.

Architecture — Role View & Modules

SwachhLens — Architecture & Data Flow

Current prototype · Firebase Spark / no-cost architecture · Human-in-the-loop



Municipal verification → resolved → citizen review / rating / reopen request

AI recommends; authorized humans decide, execute, verify and close.

Figure 2. Simplified role view for the pitch deck.

Module	Responsibility	Runs on
Citizen App	Capture, report, track, profile, feedback	Browser
Municipal Portal	Dashboard, map, alerts, dispatch, verification, supervisor workspace	Browser
Decision engines	Duplicate, priority, intervention, hotspot clustering	Browser
Gemini API	Multimodal waste classification	External API
Cloud Firestore	Shared real-time data store	Firebase Spark
Firebase Authentication	Anonymous + email/password auth	Firebase Spark
OpenStreetMap	Map tiles used by Leaflet	External tiles

Third-party / external dependencies: Google Gemini API (generateContent), Firebase Authentication, Cloud Firestore, and OpenStreetMap tiles via Leaflet/React-Leaflet.

Citizen Application

The citizen experience is mobile-first: authenticate, capture evidence, review the AI-assisted interpretation, submit, and track the case through its lifecycle.

Capability	Behavior	Status
Capture & preview	Camera capture or file selection with live preview.	LIVE VERIFIED
Compression	Client-side image resize/compression before analysis and storage.	LIVE VERIFIED
GPS + timestamp	High-accuracy browser geolocation with demo fallback; client timestamp.	LIVE VERIFIED
Reporting flow	Capture → Compress → Locate → Analyze → Review → Submit.	LIVE VERIFIED
Tracking	Human-readable complaint ID, history and lifecycle timeline.	LIVE VERIFIED
Profile	Registered profile is read-only by default with explicit Edit action; guest mode explains session limits.	LIVE VERIFIED
Feedback	1–5 rating, resolution status and reopen request.	LIVE VERIFIED

- The citizen sees the AI result before submission, so the recommendation is not a hidden system action.
- Registered citizens and anonymous guests use separate authentication paths.
- Anonymous guest data is browser-bound; clearing browser data can remove access to the guest profile.

Municipal Command Center

The municipal portal is the operational center of SwachhLens. It converts a stream of citizen complaints into a prioritized work queue, geographic view, alert center, dispatch action and verification dossier.

Capability	Behavior	Status
Dashboard	Real-time city-wide complaint metrics.	LIVE VERIFIED
Operational alerts	Urgent, high-priority unassigned, bio-hazard, aging, sensitive-location and duplicate signals.	LIVE VERIFIED
Live map	Priority markers plus current-state hotspot circles.	LIVE VERIFIED
Priority queue	Search, filters, sorting and complaint details.	LIVE VERIFIED
Dispatch modal	AI advisory beside current team load; accept or override.	LIVE VERIFIED
Verification dossier	Before/after evidence, Verify & Mark Resolved, or Send Back for Rework.	LIVE VERIFIED

Field Supervisor Workflow

Field operations are separated from city-wide control. A supervisor sees jobs assigned to the supervisor's team and updates progress from arrival through completion.

Assigned job → Arrived → Start Work → Completion photo + notes → Pending verification → Municipal Verify / Rework

- Supervisor access is team-scoped through teamId.
- Completion evidence is compressed and stored with the work record.
- The supervisor does not perform city-wide dispatch or municipal verification.

AI Pipeline

SwachhLens separates AI perception from transparent operational logic. Gemini 3.6 Flash interprets the image; client-side rules convert the structured result into priority and intervention signals that a municipal operator can inspect and override.

Attribute	Verified implementation
Model	Google Gemini 3.6 Flash (gemini-3.6-flash)
Input	Compressed Base64 JPEG + structured prompt
Output	Waste type, volume tier, confidence, location-sensitivity hint, bio-waste risk, reasoning
Categories	8: overflowing bin, garbage dump, plastic, construction debris, organic, e-waste, hazardous, drain blockage
Volumes	4: small, medium, large, very large

Failure behavior: The service validates and sanitizes malformed model output. Network/API errors are surfaced to the application rather than silently treated as successful analysis.

Priority Scoring

PriorityScore = min(100, round((Volume × 40) + (Location × 30) + (Frequency × 20) + (Age × 10) + BioRiskBoost))

Factor	Rule	Contribution
Volume	small 0.25 · medium 0.50 · large 0.75 · very_large 1.00	40%
Location	none 0.00 · main road 0.50 · sensitive areas 0.70 · drainage blockage 1.00	30%
Frequency	Nearby reports within ~50 m over 7 days, capped at five reports	20%
Age	hours since report / 48, capped at 1.0	10%
Bio-risk	+15 when bioWasteRisk is true	Boost

The portal shows human-readable reason bullets so the priority is inspectable instead of being a black-box severity number.

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Duplicate Detection & Geographic Intelligence

Duplicate detection. A candidate is retained and flagged, never deleted, when the same waste category, unresolved prior report, 48-hour window and GPS distance within 50 m are present. A 64-bit dHash adds visual similarity when an image hash is available.

Verified threshold: `DUPLICATE_RADIUS_METERS` = 50 m. The earlier 50 m / 100 m inconsistency has been resolved in favor of the current source configuration.

Hotspot intelligence. Active, non-resolved complaints are grouped into current-state spatial clusters using Haversine centroid distance with an 800 m radius. Outputs include cluster identity, centroid, report count, average priority, urgent/unresolved counts, and dominant waste type.

Important: Hotspots are real-time spatial density clustering, not predictive or time-series forecasting.

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Intervention & Dispatch

- Deterministic rules map waste attributes to manual cleanup, mini-truck support, recycling partner, drainage response and specialized hazardous response.
- Dispatch recommends an active, capability-matching team with the lowest current workload.
- The municipal operator explicitly accepts the recommendation or selects a different team/vehicle.
- Team workload changes as work is assigned and resolved.

Verification, Closure & RBAC

The final operational step is intentionally human. The operator compares the original report evidence with the supervisor's completion evidence, verifies the work or sends it back for rework, and only then closes the complaint.

Role	Primary scope
Anonymous guest	Submit reports and view own session reports; no portal access
Registered citizen	Own profile, reports and feedback
Municipal operator	City-wide command center, dispatch and verification
Field supervisor	Team-scoped execution and completion evidence

Closure loop: Before photo → after photo → municipal verification → Resolved → citizen rating / optional reopen request.

Firestore Data Model

Collection	Purpose
complaints/{id}	Incident evidence, GPS, AI result, priority, duplicate link, intervention, lifecycle, assignment and feedback.
citizens/{id}	Name, contact, area, ward, created/updated timestamps.
teams/{id}	Team type, current load, active status and zone metadata.
municipalUsers/{id}	Staff directory and role; field supervisors carry teamId.

Technology Choices & Firebase Spark Constraints

Layer	Technology	Why it is used
Frontend	React + Vite	Component-based UI and straightforward web/PWA delivery
Routing	React Router	Role-aware navigation across citizen and portal views
Auth	Firebase Authentication	Anonymous + email/password identity
Data	Cloud Firestore	Real-time operational data shared by both applications
AI	Gemini 3.6 Flash	Multimodal image understanding with structured output
Maps	Leaflet + OpenStreetMap	Interactive operational map without proprietary map infrastructure
Icons	Lucide React	Consistent UI iconography

Spark-tier decision: Cloud Storage, Cloud Functions and Cloud Run are intentionally not used. AI inference and deterministic decision engines run in the browser, while compressed images are stored as Base64 in Firestore. This keeps the prototype within the no-cost constraint but is not a production-scale storage model.

Data Ethics & Privacy

- Only limited citizen profile information is collected; contact information is not shown on public map views.
- GPS and images are collected for operational reporting and verification.
- Anonymous reporting supports low-friction participation, with the known browser-session limitation.
- AI recommendations remain advisory and municipal operators retain control over dispatch and closure.
- Production deployment should add explicit retention/deletion policies and server-side secret protection.

Assumptions & Known Limitations

Assumption / limitation	Implication
Decision support	Working prototype, not autonomous dispatch.
Client-side API key	Prototype constraint; production should move inference behind a secure backend/proxy.
Base64 image storage	Prototype-scale only; production should use object storage.
Anonymous identity	Guest sessions are browser-bound.
Video	Current input is image/photo based; video ingestion is future scope.
Hotspots	Current-state spatial clustering, not predictive forecasting.
Timestamps	Client-generated in the Spark prototype.
Maps	OpenStreetMap tiles require network access.
Deployment	Not publicly deployed and not adopted by a government body.

Future Production Architecture

- Secure server-side Gemini inference.
- Dedicated object storage for images.
- Production-grade staff authentication and role management.
- Optional video-based waste analysis.
- Richer image similarity and historical predictive analytics.
- Notifications and integrations with municipal complaint platforms.
- Automated before/after assistance as a complement to human verification.

TechNova Compliance Matrix

The current documentation evaluates 26 extracted requirement rows. Twenty-four are implemented in the prototype; two remain future scope: predictive time-series forecasting and video-stream waste extraction. This document does not claim 100% compliance.

#	Requirement	Current implementation	Status
1	Photo / visual capture	Camera/file upload + JPEG compression	PASS
2	GPS geo-tagging	Geolocation API + demo fallback	PASS
3	Timestamping	Unix-millisecond lifecycle timestamps	PASS
4	Optional comments	Complaint description field	PASS
5	AI waste classification	Gemini 3.6 Flash across 8 categories	PASS
6	AI volume estimation	4-tier volume model	PASS
7	Actionable routing	Deterministic intervention engine	PASS
8–10	Manual / heavy / recycling routing	Rule-based waste-to-resource routing	PASS
11	Hazardous escalation	Urgent flag + specialized response	PASS
12	Bio-waste detection	AI flag + priority boost	PASS
13	Drainage escalation	Drainage routing + urgency	PASS
14	Sensitive location	Location sensitivity contribution	PASS
15	Duplicate detection	Category + 50 m + 48 h + dHash	PASS
16	Explainable priority	0–100 formula + reasons	PASS
17	Operational alerts	Derived alert cards + queue filtering	PASS
18	Live GIS map	Leaflet + markers + hotspots	PASS
19	Human-in-loop dispatch	Accept / Override	PASS
20	Workload-aware routing	currentLoad recommendation	PASS
21	Field execution	Arrival → Start → completion evidence	PASS
22	Before/after verification	Human side-by-side review	PASS
23	Citizen resolution rating	1–5 stars + reopen request	PASS
24	Spatial hotspot clustering	800 m density clustering	PASS
25	Predictive forecasting	Not implemented; future scope	NOT IMPLEMENTED
26	Video-stream extraction	Not implemented; image-only prototype	NOT IMPLEMENTED

Evaluation alignment: Innovation — explainable triage, duplicate evidence, spatial intelligence and closed-loop verification. Feasibility — smartphones, existing teams, Spark-tier Firebase and a clear prototype-to-production boundary. User friendliness — role-specific flows. Ethics — role-based access, limited profile data and explicit limitations.

Conclusion & Judge-Facing Summary

SwachhLens turns a citizen waste report into verified municipal action.

The prototype captures evidence, applies multimodal AI for waste understanding, computes transparent priority and duplicate signals, supports human dispatch decisions, gives field supervisors a scoped execution workflow, and closes the loop with municipal verification and citizen review.

Evaluation dimension	How SwachhLens addresses it
Innovation & Originality	Explainable AI-assisted triage, duplicate evidence, spatial clustering, human-in-the-loop dispatch and field verification.
Technical Feasibility	React/Vite + Firebase Auth/Firestore + Gemini 3.6 Flash + Leaflet/OpenStreetMap within a Spark-tier prototype.
User Friendliness	Separate citizen, municipal-operator and field-supervisor flows with clear lifecycle and evidence states.
Data Ethics & Privacy	Role separation, Firestore rules, anonymous guest mode, limited profile data and explicit prototype limitations.

Important non-claims: SwachhLens is not production-deployed, not adopted by a municipality, not autonomous, does not analyze video, and does not provide predictive hotspot forecasting in the current prototype.

Submission package: Use this document alongside the source-code ZIP, architecture/data-flow diagrams, and the 10-slide pitch deck.